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Data Resources

RNA-KG v2.0: an RNA-centered Knowledge Graph with Properties

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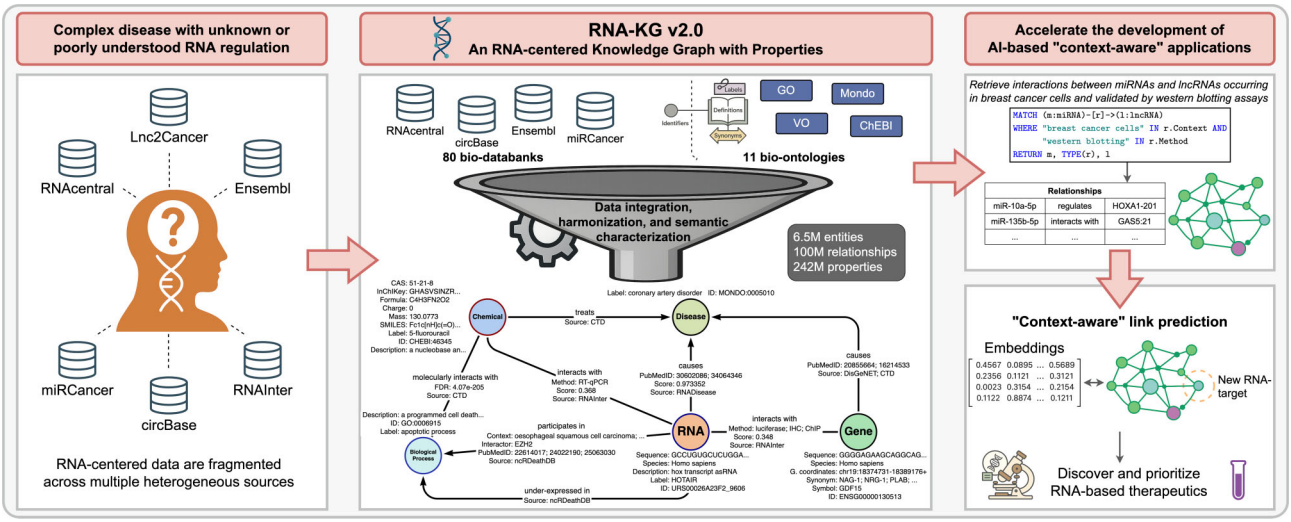
Abstract

RNA-KG is a recently developed biomedical knowledge graph that integrates the interactions involving coding and non-coding RNA molecules extracted from public data sources. It can be used to support the classification of new molecules, identify new interactions through the use of link prediction methods, and reveal hidden patterns among the represented entities. In this paper, we propose RNA-KG v2.0, a new release of RNA-KG that integrates around 100*M* manually curated interactions sourced from 91 linked open data repositories and ontologies. Relationships are characterized by standardized properties that capture the specific context (e.g. cell line, tissue, pathological state) in which they have been identified. In addition, the nodes are enriched with detailed attributes, such as descriptions, synonyms, and molecular sequences sourced from platforms such as OBO ontologies, NCBI repositories, RNAcentral, and Ensembl. The enhanced repository enables the expression of advanced queries that take into account the context in which the experiments were conducted. It also supports downstream applications in RNA research, including 'context-aware' link prediction techniques that combine both topological and semantic information. Finally, the recent integration of RNA-KG relationships into the RNAcentral portal provides a powerful resource for linking RNA-centric relationships with non-coding gene expression in human tissues, RNA secondary structures, and their functional roles in biological pathways, which can accelerate the discovery of novel therapeutic targets.

Graphical abstract



Introduction

RNA molecules play a pivotal role in various physiological and pathological processes, as confirmed by extensive research conducted over the past two decades [1–3]. Beyond their fundamental role in gene expression, these molecules have emerged as promising tools to provide new insights in the treatment of cancer, genetic and neurodegenerative disorders, cardiovascular and infectious diseases [4]. Several genomics laboratories release datasets containing interactions between RNAs and other bio-entities like genes, proteins, chemicals, diseases, and phenotypes. However, the integration and standardization of these data remain a significant challenge, since

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a common identification scheme is missing to represent RNA molecules and their relationships, and manual efforts are often required to retrieve standard identifiers.

Knowledge graphs (KGs) [5] have emerged as a powerful abstraction for integrating heterogeneous information from diverse sources, enabling the identification of complex interdependencies and the discovery of hidden relationships. While KGs were initially represented using the Resource Description Framework (RDF) [6], more recent initiatives [7–9] increasingly adopt property graphs (e.g. Neo4j [10]) due to their scalability and efficiency in managing large graphs with millions of edges. When a KG is constructed according to an ontology, the labels, types, and attributes used to describe nodes and relationships are ontology-driven, thereby ensuring consistent interpretation and semantic clarity – an essential prerequisite for supporting various reasoning tasks. In the life sciences, KGs have gained particular relevance, offering structured representations of bio-entities and their interactions, as well as supporting AI-driven methods for knowledge discovery and explanation. Several domain-specific biological KGs have recently been developed, including PrimeKG [11], the Human Disease Benchmark KG [12], the KG for hepatocellular carcinoma (KGHC) [7], ReproTox-KG [13], the Monarch Knowledge Graph [14], the Oregano Knowledge Graph [15], and the Biomedicine Knowledge Base [16], each designed to facilitate different types of analyses and research activities. However, RNA-related interactions remain underrepresented across these resources and, when included, their coverage is often sparse and incomplete. For example, the Human Disease Benchmark KG mainly focuses on coding transcripts and their localization within anatomical parts or cellular compartments, while KGHC describes only nine RNAs involved in hepatocellular carcinoma.

RNA-KG [17] is the first ontology-based KG that gathers biological knowledge about RNAs from more than 60 public databases, integrating more than 12.5*M* functional relationships with genes, proteins, chemicals, and other ontologically grounded biomedical entities. RNA-KG is a core project of the Monarch Initiative KG-Hub [18], a platform that enables standardized construction, exchange, and reuse of KGs. During the construction of RNA-KG, we have noticed that RNA-related relationships frequently depend on: (*i*) the sequence, structure, and/or the genomic loci the RNA is transcribed from; and (*ii*) the cell line, tissue, or more generally, the 'context' where they occur. Introducing this information in the KG would be an enhancement for obtaining a more sophisticated representation of the domain and could be used in several downstream applications.

This paper introduces RNA-KG v2.0, a new version of RNA-KG that has been extended from multiple viewpoints. First, relationships from 20 new data sources have been integrated, and those already stored have been more thoroughly characterized. For example, the 'generic' *RNA-causes or contributes to condition-disease* has now been detailed in specific instances (e.g. *RNA-over-expressed in-disease* and *RNA-under-expressed in-disease*) according to the metadata provided by the sources. Moreover, a new identification scheme for RNA molecules, based on RNAcentral [19, 20] for non-coding RNAs and Ensembl [21] for coding transcripts, is now adopted. This scheme enables the representation of transcript variants (i.e. isoforms), overcoming the limitations of previous releases, which collapsed multiple isoforms into a single gene-level entity. For example, the lncRNAs LINC-PINT-205 and LINC-PINT-206 previously collectively identified through the Entrez [22] identifier *Entrez:378805?lncRNA* are now represented as *URS00025E6A5F_9606* and *URS00005A4FBF_9606*. RNA entities in RNA-KG are also enhanced by integrating the Rfam [23] node classification scheme, which improves the accuracy and granularity of RNA categories. Finally, properties for both entities and relationships are now included in the knowledge base. This information is of paramount importance for discriminating bio-entities and relationships based on their data sources and the 'context' where they have been acquired. For instance, the RNA regulation of a gene may depend on specific sequence motifs. RNA isoforms may differ by a few nucleotides, but one may bind to a protein-coding gene due to the presence of the motif, while the other does not. Similarly, the over-expression of an RNA might be associated with Alzheimer's disease in brain tissues but not in the blood. This information can now be used in different kinds of analysis and for developing meaningful applications.

The resulting KG integrates entities and relationships from 80 public repositories and 11 biomedical ontologies with a total of 6,553,767 nodes involved in 99,936,712 relationships. Nodes and edges are characterized by 27,242,075 and 242,896,588 properties. The new RNA-KG version is available as a property graph in Neo4j. A web portal and an API support the users in accessing and interacting with the information stored in the knowledge base. By means of the web portal, users can decide to generate *views* (i.e. subgraphs of interest [24]) by exploiting the endpoint or download *views* already extracted by our team. Thanks to the adoption of standard identification schemes, the integration with the information of other portals is made easy, and further kinds of analysis can be carried out. This is also demonstrated by the recent integration of RNA-KG relationships into the RNAcentral portal [19].

By exploiting the new release of RNA-KG and its API that provides easy access to entities and relationships with their properties, a researcher can: (*i*) extract subgraphs of interest via Cypher [25] queries, for example selecting interactions validated by specific experimental methods (e.g. qPCR, western blotting) or RNAs with particular properties (e.g. GC content above a given threshold) or sharing the same functional annotations; (*ii*) apply multimodal machine learning (ML) approaches that combine information from RNA-KG, such as sequence and topological features, with external data sources (e.g. RNA editing events like A>I from REDIportal [26] or tissue-specific expression profiles from Expression Atlas [27], tools both integrated into RNAcentral), enabling context-specific analyses of RNA functions and interactions; (*iii*) support hypothesis generation and prioritize candidate RNA molecules using ML (e.g. inferring RNAs plausibly associated with a disease).

The paper is organized as follows. The Materials and Methods section introduces the new data sources, the main modifications to the identification scheme, and the characteristics of the bio-entities, relationships, and their properties. An excerpt of the KG is also presented to show its expressive power. The Result section describes the web portal, the API, and several use cases and applications in which RNA-KG can be fruitfully used. The Discussion section provides concluding remarks and future research directions. Finally, the Data Availability section details how to access the web portal, the API, and the code

of the use cases. The code is available in our public repository to support the reproducibility of the experiments.

Materials and methods

New data sources and identification schemes

RNA-KG v2.0 integrates publicly available relationships involving biomedical entities from 80 data sources and 11 ontologies. Twenty new sources were identified: RNAcentral, Ensembl, DisGeNET [28], GeneMANIA [29], CTD [30], ClinVar [31], STRING [32], starBase2 [33], microT [34], The Human Protein Atlas [35], miRanda [36], Reactome [37], HGNC [38], GTEx [39], POSTAR2 [40], UniProtKB [41], RNAhybrid [42], PhenomiR [43], COSMIC [44], and circBase [45]. These sources were considered because they (*i*) increase the coverage of interactions that describe the 'RNA world' (e.g. *RNA-ribosomally translates to-protein* interactions from Ensembl); (*ii*) are well-reputed sources that do not or marginally deal with RNA molecules but enhance the KG quality, including relationships involving entities that are connected to RNA molecules in RNA-KG (e.g. *protein-molecularly interacts with-protein* interactions from STRING). Supplementary Tables S1–S3 detail the new relationships included along with the entities involved.

Eleven standard biomedical ontologies are used to establish common semantics: Human Phenotype Ontology (HPO [46]), Gene Ontology (GO [47]) Monarch Merged Disease Ontology (Mondo [48]), Vaccine Ontology (VO [49]), Chemical Entities of Biological Interest (ChEBI [50]), Uber-anatomy Ontology (Uberon [51]), Cell Line Ontology (CLO [52]), PRotein Ontology (PRO [53]), Sequence Ontology (SO [54]), Pathway Ontology (PW [55]), and Relation Ontology (RO [56]). We selected these ontologies because their terms, properties, and hierarchical structures are widely accepted by the scientific community to describe biological classes and entities, such as diseases, phenotypes, chemicals, biological processes, proteins, and relations between them, consistently and unequivocally. Bio-entities that are not currently represented in one of these ontologies are modeled using well-accepted terminologies. They include DrugBank [57] accession numbers for identifying drugs, NCBI Entrez identifiers for genes and pseudogenes, RNAcentral identifiers for ncRNA molecules, Ensembl transcript identifiers for mRNA sequences, NCBI dbSNP [58] rsIDs, and COSMIC identifiers for germline and somatic variants. Grounding RNA molecules in RNAcentral and Ensembl identifiers allows the precise RNA sequence to be distinguished among RNA isoforms. Table 1 details the identification schemes for the main bio-entities according to the proposed bar charts.

The acquisition of raw data and its transformation into entities and relationships in RNA-KG v2.0 followed the same pipeline adopted in [17] and leveraged the primitives provided by the PheKnowLator library [12]. Specifically:

- Tabular data were downloaded from the considered data sources, cleaned, and semantically annotated. The semantic annotation serves two main purposes. First, it identifies the entities and the relationships (with their properties) to be included in the KG according to the structure of the classes of the previously mentioned ontologies and the relationships of the Relation Ontology. Second, it specifies when to apply lookup tables to convert local data source identifiers into the identification schemes used in RNA-KG.
- Then, mapping rules were applied to each row of the tabular data to generate the bio-entities and their relationships. These rules exploit lookup tables and also exclude tuples that do not meet quality constraints (e.g. relationships with high p-values or other reliability metrics).
- By leveraging the Neo4j import utilities, the graph database was generated from the cleaned ontology-grounded data. A Neo4j endpoint was created to make RNA-KG v2.0 freely available and accessible. Using Cypher, portions of the graph can be extracted to perform 'context-aware' analyses that address specific biological questions.

Table 1. Identification schemes for the main bio-entities

Entity	Scheme	Example
Variant	dbSNP COSMIC	rs766102409 COSV60127483
ncRNA	RNAcentral	URS00000478B7_9606
mRNA	Ensembl	ENST00000713680
circRNA	circBase	hsa_circ_0018046
Chemical	ChEBI	CHEBI:4021
Protein	PRO	PR:Q8WXF3
Gene	Entrez	Entrez:1954
Cell	CLO	CLO:0003725
GO term	GO	GO:0140657
Disease	Mondo	MONDO:0020683
Phenotype	HPO	HP:0040064
Anatomy	Uberon	UBERON:0002169
Vaccine	VO	VO:0010137
Pathway	PW Reactome WikiPathways	PW:0000035 R-HSA-9837999 WP5090

RNA-KG bio-entities

Node type distribution

RNA-KG includes approximately 6.5*M* typed nodes with 139 distinct node types. Ontology classes are used to define node types (e.g. *SO:0000276* for `miRNA`). Additional node types have also been introduced by parsing ontologies to enhance entity types with specific relevance within the 'RNA world'. For instance, the type `Cardiovascular disease` was assigned to specific disease entities based on hierarchical relationships within Mondo, and the `Transcription Factor` (`TF`) type has been associated with proteins whose label in PRO contains the string 'transcription factor' or matches known TF symbols from curated sources. Moreover, the `GO` type is further split into `Cellular component`, `Molecular function`, and `Biological process`, consistent with the three branches of the Gene Ontology. This subdivision is particularly relevant as it enables a focus on biolog-

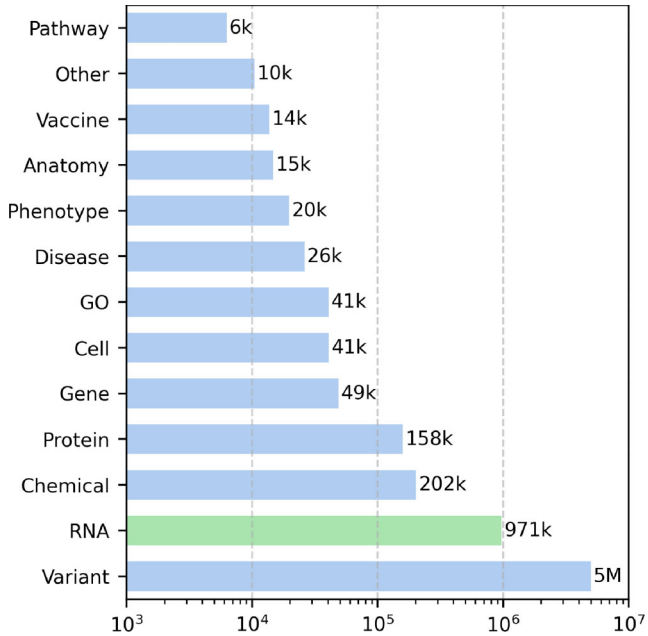


Figure 1. Node type distribution.

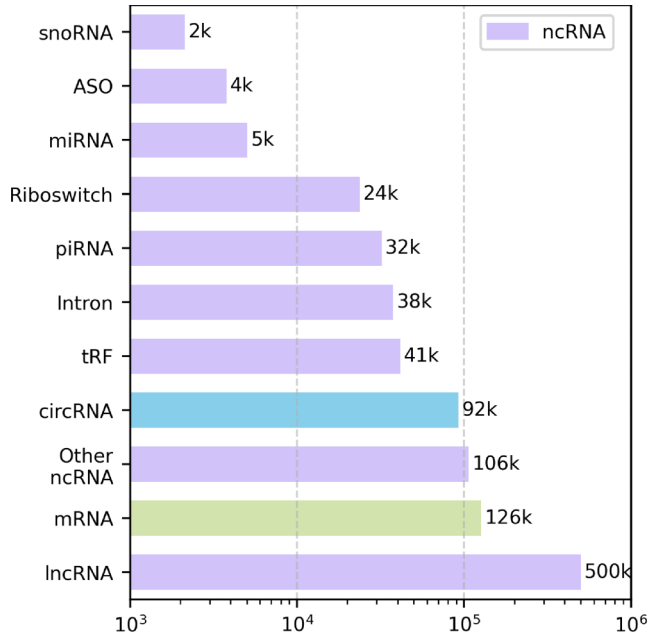


Figure 2. RNA node distribution.

ical processes (such as *A-to-I tRNA editing*) that are extensively studied in the context of RNA biology. Finally, this fine-grained representation facilitates the construction of domain-specific *views* using Cypher queries or the graphical interfaces of the web portal.

Besides the well-represented entities, RNA-KG also incorporates less frequent ones, including hundreds of `Protein modifications` (e.g. *hydroxylated lysine*), `Epigenetic modifications` (e.g. 3-(3-amino-3-carboxypropyl)uridine – *acp3U*, from MODOMICS [59]), and `Histone modifications` (e.g. *H3K27 monomethylation site*). The relevance of this knowledge is increasingly recognized in both physiological and pathological processes involving RNA molecules and can be used to investigate RNA-mediated mechanisms [60].

Figure 1 shows the node types distribution in RNA-KG. The most represented class is `Variant` (with around *5M* nodes) due to the integration of large-scale variant databases (e.g. dbSNP and COSMIC) that catalog millions of mutations across the human genome. Note that each known mutation event (e.g. an A-to-T substitution occurring in a specific genomic locus) leads to the creation of a new variant identifier. The second most common node type is `RNA` (971*k* nodes). RNA sequences are typed using SO classes and Rfam categories such as `mRNA`, `lncRNA`, and `miRNA`. Figure 2 details the distribution of RNA types. Non-coding RNAs represent a substantial portion of this set (around 752*k* nodes), as supported by current estimates [20, 23]. mRNAs account for around 126*k* nodes, a number consistent with the expected diversity of transcript isoforms derived from the 20*k*–22*k* protein-coding genes in the human genome [61].

Bio-entities' properties

Properties for RNA-KG nodes correspond to the attributes associated with the ontological terms. Among the available attributes, `label`, `description`, and `synomyms` are associated with nodes that represent amino acids, anatomical parts, biological processes, biological roles, cells and cell lines, cellular components, chemicals, diseases, genomic features such as epigenetic and histone modifications and sequence locations, protein modifications, lipids, molecular functions, pathways, phenotypes, proteins, species, and vaccines. Additional attributes are included for specific nodes' categories. Specifically, `charges`, `masses`, `formulae`, `SMILES structures`, `CAS numbers`, and `InChIKeys` for chemical entities; amino acid `sequences` and `species` for proteins.

On the other hand, RNA-KG nodes representing entities that cannot be directly mapped to ontological terms (e.g. genes, RNAs, and variants) are characterized by a subset of the attributes that can be retrieved from the corresponding terminologies. Specifically, (*i*) DrugBank provides `labels`, `sequences`, `synonyms`, `InChIKeys`, and `CAS numbers` for drugs; (*ii*) NCBI Entrez provides `labels`, `species`, `synonyms`, `descriptions`, and `genomic coordinates` with associated nucleotide `sequences` for genes; (*iii*) NCBI dbSNP and COSMIC provide `labels`, `species`, and `genomic coordinates` with associated `mutations` that occur in the nucleotide chain. Many efforts have been devoted to producing a uniform representation of specific attributes like `structures`, `sequences`, `mutations`, and `genomic coordinates`. Figure 3 shows the distribution of nodes' properties in RNA-KG, highlighting the cardinality of RNA nodes' properties, which will be detailed in the following.

Properties specific for RNA molecules

RNA molecules are characterized by a set of properties (e.g. `labels`, `structures`, `descriptions`, `genomic coordinates`, and `sequences`) that are mainly acquired from centralized repositories like Ensembl and RNAcentral, minimizing the risk of heterogeneity in annotation. The property `genomic coordinates` is the most numerous since it is a list of genomic loci (whose items are provided with the standard format *chromosome: start-end strand*) from which RNA molecules are transcribed (the same RNA sequence can present multiple loci in the genome). Proprietary specialized databases contribute additional annotations when molecules

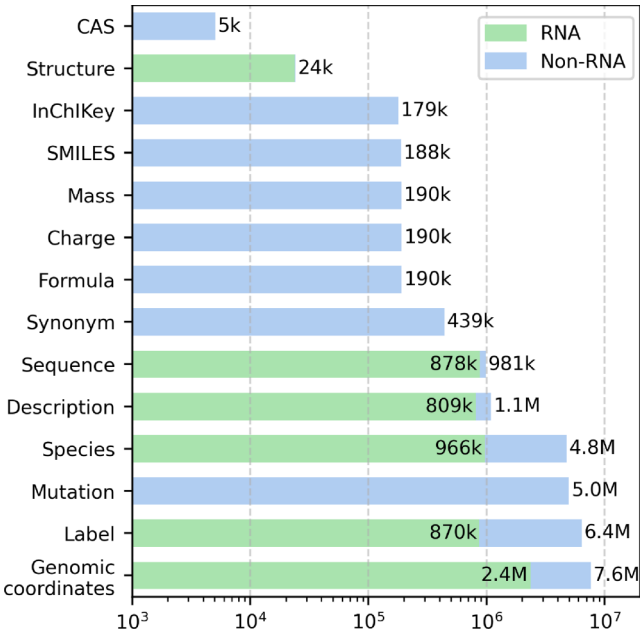


Figure 3. Node property distribution.

cannot be mapped on Ensembl/RNAcentral terminologies (details in the Supplementary material).

Figure 4 shows the hierarchical organization of RNA molecules along with the associated properties. A property is added to an RNA class if at least one instance of that class is enriched with that property. Child nodes inherit parent nodes' properties. For the sake of readability, RNA drugs and many RNA (sub)categories that are under-represented in RNA-KG are removed from the visualization. For example, RNA antisense oligonucleotides subclasses (e.g. PMO ASO, 2MeO ASO), tRF subtypes (e.g. tRF-1, tRF-3), scaRNAs, T-box riboswitches, ribozymes, and RNA components of RNAses, retained introns, precursor RNAs, bacterial RNAs, and RNA decay molecules are removed because they do not add relevant properties to the hierarchy. It is worth noting that nucleotide sequences are always associated with an RNA class, but some of them miss the associated genomic coordinates because they are synthetic (e.g. siRNAs, shRNAs, ASOs) or they are co-located with multiple transcripts, as for piRNAs, and no certain coordinates are provided by the original source. Viral RNAs (that usually represent the entire genome of the pathogen since RNA virus genomes usually contain relatively few genes [62]), bacterial riboswitches, and tRNA sequences are enriched with their secondary `structures` (expressed using the base pair 'dot-bracket' notation [63]). These foldings are crucial for RNA functional analysis as they influence RNA stability, interactions with proteins/RNAs, and the regulation of gene expression [64].

RNA-KG edges

Edge type distribution

RNA-KG includes approximately 100*M* typed edges with 424 distinct edge types. Edge types are grounded in OBO ontologies' properties. Specifically, we used RO properties for annotating relationships identified within linked open data sources. The hierarchical organization of concepts in RO allows the expression of different kinds of edges at different granularities. For example, the general property *RO:0002434 interacts with* (the most numerous edge type in RNA-KG due to the integration of homogeneous sources such as RNAInter [65]) can be substituted with more specific properties such as *molecularly interacts with* or *genetically interacts with*. Inverse relationships are included in RNA-KG (e.g. *location of* and its inverse *located in*). Besides edges coming from linked open data, we introduced edges stemming from (*i*) the integration of the eleven bio-ontologies according to the abstraction strategy described in [66] and (*ii*) the entity linking technique according to the class-based modeling described in [12]. The model links entities not directly mappable to an ontology term (i.e. database entities without a corresponding ontological identifier) to ontology classes using the *rdfs:subClassOf* property.

Figure 5 shows the distribution of edge types in RNA-KG, highlighting the cardinality of the relations involving RNA molecules. The distribution mirrors the biological focus of RNA-KG since more than half of its relationships involve RNAs. The figure also points out the relevance of *rdfs:subClassOf*, which is used for entity linking and ontology integration.

Edges' properties

Figure 6 outlines the most important edges' properties that we have identified in terms of cardinality and biological relevance, highlighting properties involving RNA molecules. Examples of properties are (*i*) the `score` that indicates the level of confidence for the interaction to occur; (*ii*) the `method` that provides experimental evidence to the interaction (e.g. RT-PCR, AGO-CLIP, ImmunoBlot); (*iii*) the specific tissue, cell line, disease, phenotype (in general, the `context`) where the interaction was analyzed; (*iv*) the `PubMed identifier` of the article where the interaction is described. Interactions can also be enhanced by `variants`, `epigenetic modifications`, competing molecules or binding partners (RNAs, genes, etc., in general, the `interactor`) that provides further context for the documented relationship as described by the original source. Each relationship within RNA-KG is also associated with the list of `sources` providing the interaction. Moreover, edges' properties are grounded in bio-ontologies and terminologies, as we have done for nodes and edge types. For example, `methods` are grounded in the National Cancer Institute Thesaurus (NCIt [67]), `mutations` in dbSNP and COSMIC, and `regulators` in PRO.

Since we are aware that binary relationships usually adopted in the representation of KG are not adequate to model dependencies between properties (e.g. the dependence of mutations on the context) and ternary relationships (e.g. those observed in ceRNA interactions [68]), we have chosen to represent this information as a list of attributes because it offers a practical compromise between expressivity and simplicity. Modeling these complex dependencies explicitly would require blank nodes or reification patterns, which would increase both the complexity of the graph schema and the computational overhead in querying and maintenance. Instead, through the proposed representation, we can easily extract subgraphs containing the required information for conducting a specific type of analysis, thereby simplifying the application of ML approaches, especially those that are graph-based.

Example 1. *To illustrate the expressive power of the property-graph edition of RNA-KG, consider the subgraph in Fig. 7 depicting the interactions among the miRNA* `hsa-let-7a-5p`*, the protein-coding gene* `CDC34`*, and the small compound* `metformin` *(for simplicity, labels are used instead of identifiers). Node properties provide biological information to be*

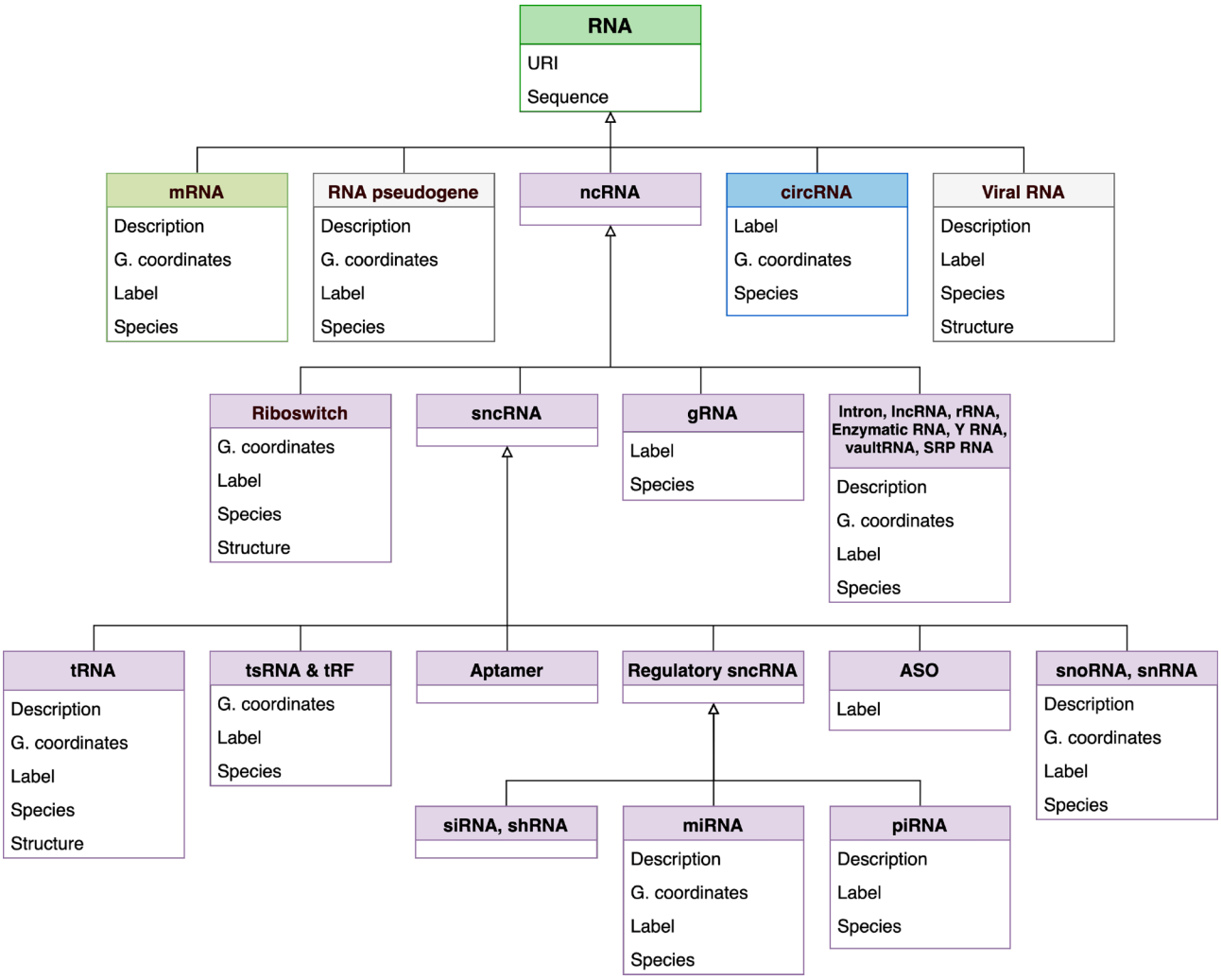


Figure 4. Hierarchy of RNA molecules with associated properties.

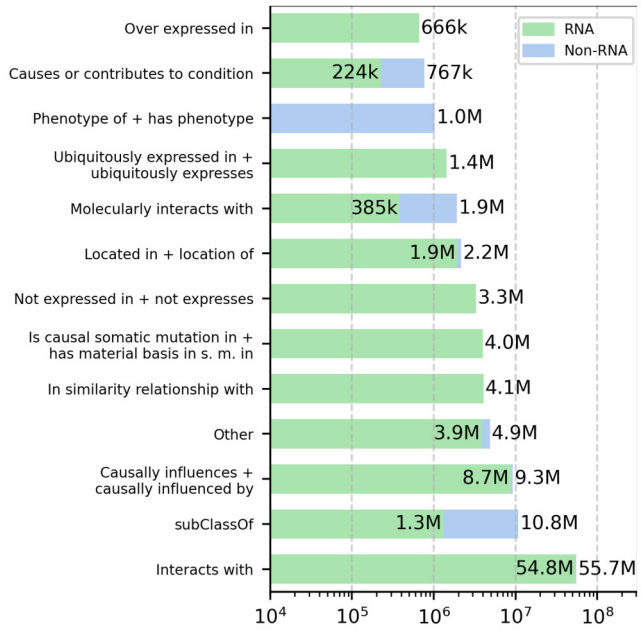


Figure 5. Edge type distribution.

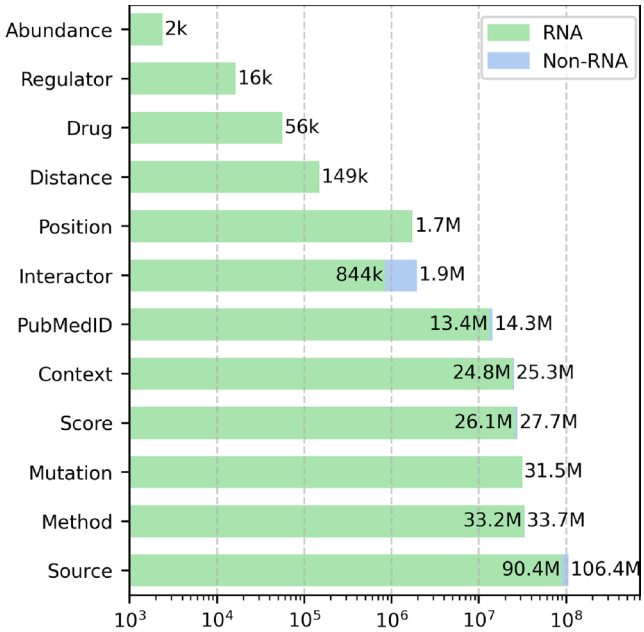


Figure 6. Edge property distribution.

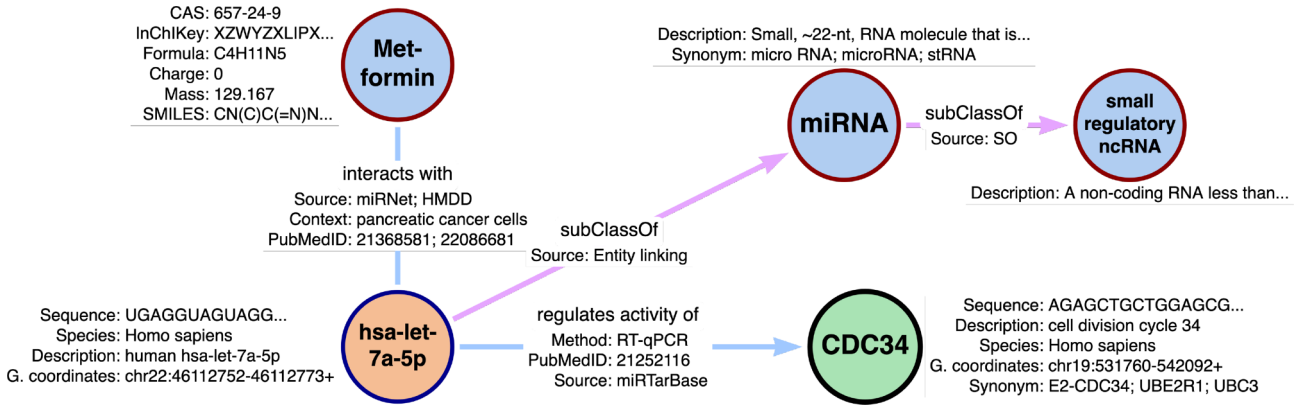


Figure 7. Excerpt of an RNA-KG subgraph.

exploited in downstream analyses. `Species` *and* `description` *are associated with miRNAs and genes and facilitate the disambiguation, curation, and filtering of nodes by organism or functional role.* `Genomic coordinates` *and* `sequence` *can support sequence-based analyses such as RNA-target binding prediction, isoform comparison, and integration with data coming from omic studies. Metformin is annotated with attributes derived from the ChEBI ontology, including* `formula`, `SMILES`, `mass`, *and* `charge`. *Combining these features with topological metrics (e.g. node degree, centrality, closeness) enables chemo-informatics analyses that integrate chemical and network-based evidence to prioritize molecules.*

Edge properties enrich the semantics of interactions. The edge regulates activityof *between* `hsa-let-7a-5p`, *and* `CDC34` *includes the properties* `method`, `PubMedID`, *and* `source`. *These attributes assess the reliability of interactions and can be used in interaction-ranking applications* [69]. *The edge* `hsa-let-7a-5p` interacts with `metformin` *presents the* `context` *(i.e. the cancer cells where the interaction was observed) and the* `sources` *confirming the interaction. During views generation, this knowledge can be used to weigh interactions and prioritize them.*

RNA-KG v2.0 also integrates entity linking and ontology nodes and edges (pink relations). `hsa-let-7a-5p` *is linked to its corresponding SO class* `miRNA` *via a* rdfs:subClassOf *relation. This allows hierarchical querying in Cypher and the retrieval of entities based on their ontology categories. The SO hierarchy includes links between ontology terms (e.g.* `miRNA` *and* `small regulatory ncRNA`*) that can be used for reasoning tasks. Furthermore, ontology nodes' properties (e.g.* `label`, `description`, *and* `synonyms`*) facilitate entity normalization, mapping, and semantic search.*

Results

The Web portal and API

A web portal is available at https://RNA-KG.biodata.di.unimi.it and provides several facilities for working with RNA-KG v2.0. As shown in Fig. 8(a), the portal provides a browsing interface for exploring the KG structure. Starting from a node of the graph, it is possible to expand the visualization with its neighboring nodes and see the properties associated with nodes and edges. Moreover, through the interface in Fig. 8(b), the user can generate a customized *view* by specifying the kind of nodes and edges to be included. Customized and predefined

```
{
    "node_uri": "https://www.ensembl.org/Homo_sapiens/Transcript/
     Summary?t=ENST00000337147",
     "node_id": "Ensembl:ENST00000337147",
     "node_rnakg_id": 537604,
     "node_labels": ["RNA", "mRNA"],
     "node_properties": {
           "Description": "ezrin",
           "Label": "EZR-201",
           "Genomic_coordinates":["chr6:158765741-158818227-"],
           "Sequence": "GGCGCGGCAGGAGG...",
           "Species": "Homo sapiens"
       }
}
```

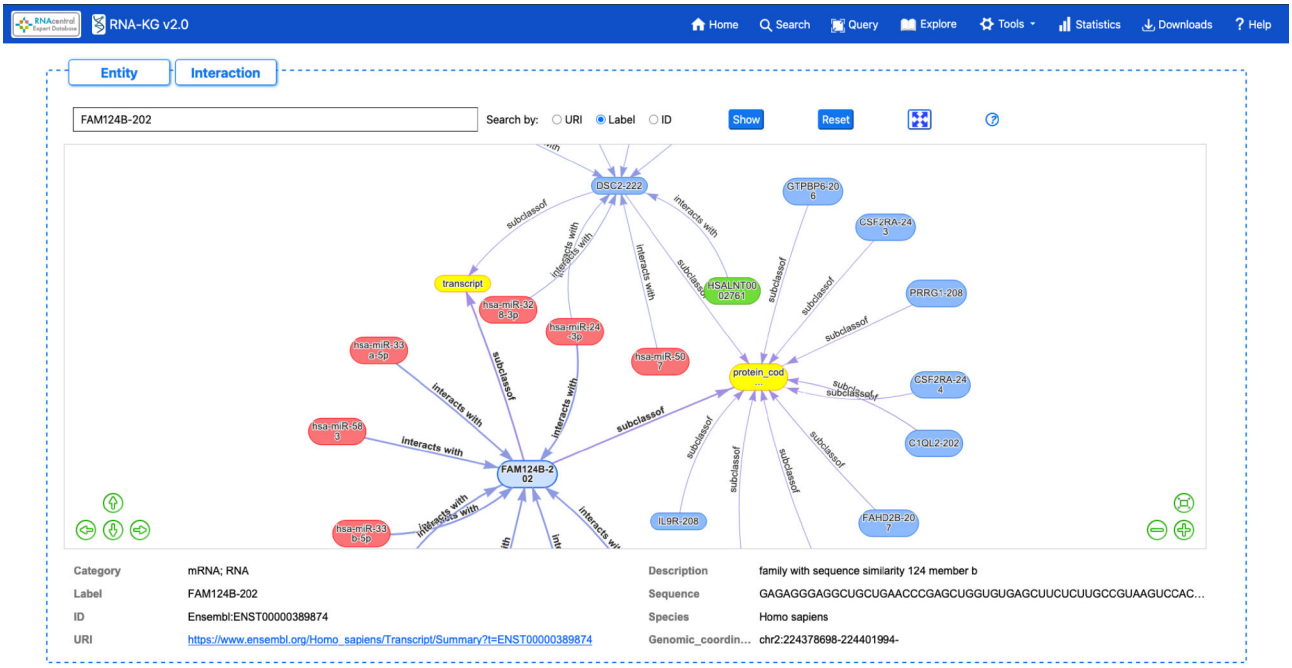
Listing 1. API JSON response for the EZR-201 transcript.

views (prepared by our team) can also be downloaded in CSV format from the web portal. RNA-KG v2.0 is also accessible via a RESTful API. The API exposes endpoints to retrieve the properties of a given entity, incoming and outgoing relationships for an entity, to run Cypher queries on the Neo4j database, and to access metadata and schema-level information. It also supports filtering options (e.g. for RNAcentral-derived relationships or for including/excluding specific fields) and provides JSON-formatted responses, making it easier to integrate with other tools.

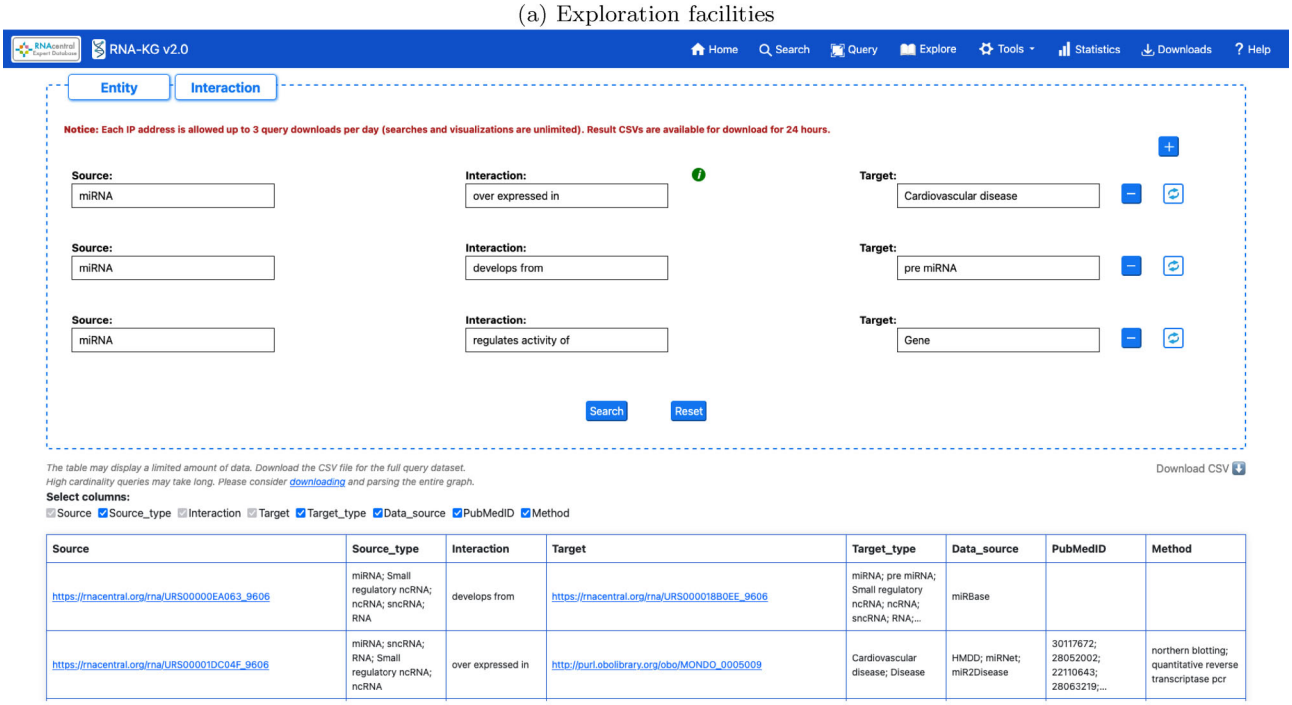
Example 2. *Suppose we are interested in the Ensembl transcript* `EZR-201` *(with identifier ENST00000337147). The following API can be invoked to obtain the JSON in* Listing 1 *with the genomic coordinates, description, and sequence.* https://rna-kg.biodata.di.unimi.it/api/v1/node/id?node_id=ENST00000337147&node_id_scheme=Ensembl

The RNA-KG v2.0 API also enables the retrieval of topological information, sequence-based evidence, and node- and edge-level properties of an RNA molecule to prioritize the most promising candidates for experimental validation.

Example 3. *The following API call makes easy to prioritize the miRNAs that down-regulate the transcript* `EZR-201`. https://rna-kg.biodata.di.unimi.it/api/v1/incoming/id?node_id=ENST00000337147&node_id_scheme=Ensembl



(a) Exploration facilities



(b) Generation of custom *views*

Figure 8. RNA-KG web portal.

Among the different relationships in which `EZR-201` *can be involved that are contained in the JSON response, we can extract those of type* directly negatively regulates activity of, *which corresponds to the canonical repression mechanism exerted by miRNAs on mRNAs (as also formalized in the RO ontology term). Four candidate miRNAs downregulate* `EZR-201` *(among the more than 5,000 miRNAs documented in RNA-KG):* `hsa-miR-183-5p`, `hsa-miR-335-3p`, `hsa-miR-205-5p`, *and* `hsa-miR-345-5p`.

A list of the main endpoints with their descriptions, input/output examples, and HTTP methods is provided in Supplementary Table S4. The complete API documentation is available at https://RNA-KG.biodata.di.unimi.it/api/v1/docs, with details of usage and outputs.

Applications and use cases

RNA-KG v2.0 enhances its utility in multiple ways. First, it enables fine-grained queries, allowing users to filter interactions not only by molecular entities but also by context, experimental method, or supporting literature. Furthermore, node and edge properties can also be leveraged as additional features to improve the performance of ML models across different

```
MATCH (m:miRNA)-[r]->(g:Gene)
WHERE "western blotting" IN r.Method
RETURN m.Label AS miRNA, TYPE(r) AS rel_type, g.Label AS Gene
```

miRNA	rel_type	Gene
hsa-miR-210-3p	regulates activity of	MIR210HG
hsa-miR-145-5p	regulates activity of	NKX2-1-AS1
hsa-miR-25-3p	regulates activity of	CBR3-AS1

Listing 2. Query to retrieve interactions between miRNAs and genes validated by a western blotting assay in RNA-KG (and excerpt of the result).

```
MATCH (m:miRNA)
WITH m, REDUCE(u_count = 0, c IN SPLIT(m.Sequence, "") |
     u_count + CASE WHEN c = "U" THEN 1 ELSE 0 END) AS u_total,
     SIZE(m.Sequence) AS total
WHERE 100.0 * u_total / total > 55
RETURN m.Label AS miRNA, m.Sequence AS Sequence
```

miRNA	Sequence
hsa-miR-4775	UUAAUUUUUUGUUUCGGUCACU
hsa-miR-3613-5p	UGUUGUACUUUUUUUUUUGUUC
hsa-miR-4753-3p	UUCUCUUUCUUUAGCCUUGUGU

Listing 3. Query to retrieve miRNAs whose uracile percentage is above 55% (and excerpt of the result).

tasks. Finally, bio-entities interactions can be integrated with information from external sources (e.g. RNAcentral) via the RNA-KG API, enabling further analysis. In the remainder, different applications and use cases will be discussed. The corresponding code is available in the `experiments` directory of the GitHub repository.

Context-aware Cypher queries

Cypher can be exploited for extracting node properties, direct and indirect relationships among bio-entities, and subgraphs. Several conditions can be easily expressed depending on the research questions the biologists have in mind.

Example 4. *Suppose biologists are interested in identifying the interactions between miRNAs and genes that western blotting assays have experimentally validated.* Listing 2 *shows the Cypher query and an excerpt of the corresponding result to retrieve relationships between miRNAs and genes. Whenever the relationship* `r` *has been confirmed by western blotting assays, the query returns the label of the miRNA molecule, the type of interaction, and the label of the target gene. By its execution, we discover that* regulates activity of *is the only relationship type between them. Furthermore, the query retrieves 2,738 interactions involving 651 miRNAs and 1,032 genes, confirming that a single miRNA can regulate multiple target genes* [1]. *Only mature miRNAs are involved as they are incorporated into the RISC complex and are functionally active in the recognition and regulation of target genes.*

Example 5. *Suppose biologists wish to retrieve miRNAs that are rich in uracil, which may be relevant for structure, stability, and molecular network studies. This request requires imposing conditions on the miRNA sequences.* Listing 3 *shows the corresponding query and an excerpt of the result. We first determine for each miRNA molecule the* `total` *number of uracil bases contained in its sequence (using the* `WITH` *clause). Then, miRNAs' label and sequence whose uracil ('U') content exceeds 55% are returned.*

In RNA-KG, many of these uracil-rich miRNAs are connected to specific transcription factors and regulate genes. For example, `hsa-miR-3613-5p` *interacts with metal-response element-binding transcription factor 2 (*`MTF2`*) while also regulate a large set of ribosomal protein genes, translation factors, and pseudogenes, including* `RPL26P6`, `RPS4XP22`, `ETF1P3`, `EIF4EP2`, `UBE2D3P1`, *and* `MYL6P5`. *This pattern suggests that uracil-rich miRNAs may preferentially interact with TFs with a primary role in histone and repressor binding activity (such as* `MTF2`*) and regulate genes involved in ribosomal and pseudogene networks. Associations between miRNA sequences and the functional pathways of their targets have been reported in* [70], *indicating that GU-rich motifs are often involved in specific regulatory mechanisms.*

```
MATCH (m:miRNA)-[rmg:regulates_activity_of]->(g:Gene)-
    [rgd:causes_or_contributes_to_condition]->(d:Disease)
WITH m, g, d,
     [pmid IN rmg.PubMedID WHERE pmid IN rgd.PubMedID] AS
     common_pmids
WHERE SIZE(common_pmids) > 0
RETURN m.Label AS miRNA, g.Label AS Gene,
    d.Label AS Disease, common_pmids AS PubMedID
```

miRNA	Gene	Disease	PubMedID
hsa-miR-34a-5p	NAMPT	inherited obesity	[23834033]
hsa-miR-200c-3p	ZEB2	breast carcinoma	[19839049]
hsa-miR-200b-3p	ZEB2	breast carcinoma	[19839049]

Listing 4. Query to retrieve miRNA-gene-disease triples with shared PubMedIDs (and excerpt of the result).

Example 6. *Suppose biologists are interested in identifying PubMed articles that describe how miRNAs regulate the activity of genes known to cause or contribute to the development of a specific disease. In this case, we need to identify a path of length 2 in RNA-KG where the first entity is a* `miRNA` *that is connected through a relationship* regulates activity of *with a* `Gene` *that in turn is connected through a relationship* causes or contributes to condition *with a* `Disease`. *When this pattern is identified, the common PubMed articles that support both interactions are retrieved.* Listing 4 *shows the corresponding query and an excerpt of the result. The* `MATCH` *clause reports the identification of the path among* `miRNA`, `Gene`, *and* `Disease`. *Then, the* `WITH` *clause computes the intersection of PubMed IDs between the two relationships for each miRNA–gene–disease triple. Only ternary relationships with at least one shared PubMed ID are returned, along with the miRNA, gene, disease, and supporting PubMed IDs.*

From the execution of this query, we observed that: (i) in PubMed article 23834033, *the miRNA* `hsa-miR-34a-5p` *is associated with inherited obesity when it regulates the gene* `NAMPT`, *(ii) in PubMed article* 19839049, *the miRNAs* `hsa-miR-200c-3p` *and* `hsa-miR-200b-3p` *regulate the gene* `ZEB2`, *contributing to breast carcinoma.*

Thus, RNA-KG 2.0 allows: (i) given a PubMed article, to identify relationships without having to read the full text, and (ii) given specific relationships (e.g. miRNA–gene–disease associations), to determine the articles supporting them.

All these queries allow the extraction of tabular data that can be used for conducting different kinds of analyses. They

can be used to train supervised ML techniques or to group data according to specific hypotheses or research questions.

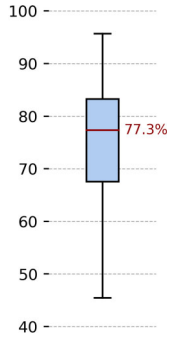
Content-aware KG pruning and clustering

Traditional approaches to KG pruning (i.e. the process of conflating or removing nodes and edges that are irrelevant to a specific analysis [71]) and clustering (i.e. grouping highly similar nodes) typically rely on topological criteria [72, 73] or incorporate 'domain-aware' knowledge [74, 75]. For example, the GRAPE [76] library identifies 'topological oddities' of a graph, that is, nodes that are topologically indistinguishable because they share the same type and neighbors. Moreover, nodes can be clustered together to discover communities relying on connected components, clustering coefficients, and neighborhood similarity, as well as low-dimensional projections via dimensionality reduction techniques like t-SNE. However, in many use cases, it is desirable to complement structural information with the semantic content encoded in the node and edge attributes. For example, 'biologically similar' RNA molecules can be clustered by considering sequence alignment techniques [77, 78] or according to the tissue in which they are expressed or active, allowing the discovery of tissue-specific and functionally-related entities.

In all these situations, the rich set of properties that are made available in RNA-KG v2.0 can be exploited to define approaches that are both topological and content-aware.

Example 7. *Suppose we are interested in studying physical interactions involving sncRNAs. The subgraph of interest consists of approximately 9.8k nodes, 244.8k edges, and 4 edge types:* interacts with, genetically interacts with, molecularly interacts with, *and* directly regulates activity of. *In this subgraph, we identified 177 isomorphic node groups involving 2.8k nodes (mainly miRNA and tsRNA sequences) and 139.3k edges (56.9% of the total). This high density can be explained by the biological tendency of tsRNAs to act in coordinated groups* [79]. *A topological pruning approach would collapse all nodes in each group into a single representative, potentially reducing the edges to 48k. Although this strategy would greatly reduce the size of the graph, it would also risk discarding biologically significant differences among the sequences.*

To mitigate this phenomenon, the average global sequence alignment scores can be computed within each group using the Needleman–Wunsch algorithm [78]. *Fig. 9 shows the distribution of alignment scores across groups. A total of 78 groups (49.7%) involving 596 sequences exhibit a score exceeding the median (highlighted in red in the boxplot). By only collapsing nodes within groups whose members show high biological similarity (i.e. groups with alignment scores above the median), the graph size is reduced to 9.3k nodes and 196.3k edges (a 19.8% reduction) while preserving both topological and content-aware information.*



Alignment score (%)	Groups	Seq.
>90	19	72
90–80	47	386
80–70	42	834
70–60	36	1,182
60–50	11	153
<50	2	5

Figure 9. Alignment scores distribution of sncRNA isomorphic groups.

Topological and context-aware link predictions

Link prediction is a core task in KG analysis, aiming to infer new plausible edges between entities based on the graph structure and content. Existing approaches only exploit the network topology (homogeneous methods), the topology and the nodes' and edges' types (heterogeneous methods), or that also combine the node and edge properties (multimodal methods).

We have recently demonstrated [24] that when interested in identifying the existence of an edge between two KG nodes, simple homogeneous graph representation learning (GRL) methods (e.g. LINE [80] and node2vec [81]) combined with Random Forests (RF) achieve a balanced accuracy above 80% for homogeneous link prediction tasks involving miRNA-miRNA, miRNA-gene, and lncRNA-disease relationships. Thus, these approaches can be used when we are interested in the existence of a link or when the KG is quite homogeneous and a few types of edges occur between the same pair of node types.

When we wish to identify the existence of an edge between two KG nodes and the type of their relationship, heterogeneous GRL methods (e.g. TransE [82], ComplEx [83], and Metapath2Vec [84]) can be employed. These methods embed both node and edge types, thus predicting the existence of a link and the specific interaction type. These methods are relevant when we need to discriminate among multiple semantic relations that may exist between the same types of entities. For example, when studying RNA–disease interactions, we might be interested in determining whether an RNA molecule is *over-expressed in*, *under-expressed in*, *contributes to*, or *causes* a given disease.

Multimodal link prediction methods [85] further extend homogeneous and heterogeneous approaches by incorporating complementary information contained in nodes' and edges' attributes in the learning process. For instance, fine-tuned transformers like BioBERT [86] or DNABERT [87] can be used to embed textual biomedical descriptions and sequences. These embeddings can be concatenated with those obtained by GRL methods to enhance the topologically based prediction method with biological and contextual information.

Example 8. *To evaluate the behavior of the link prediction approaches on RNA-KG, we generated a* view, *named* `miRNAdisease`, *with 105k nodes and 1.6M edges that aims to represent the biological and functional context of miRNAs by including their interactions with diseases, phenotypes, epigenetic changes, and genes. miRNAs and genes are enhanced with sequences, while diseases and phenotypes are enhanced with ontological descriptions. Supplementary Fig. S1 reports details on node distribution and edge cardinality (only edges with more than 1k occurrences are shown for the sake of readability).*

Table 2 reports the balanced accuracy on our view. We distinguish between homogeneous link prediction (denoted `Hom`, *predicting the existence of an edge, regardless of its type), het-*

Table 2. Link prediction results on *miRNAdisease*

Model	Edge	Balanced accuracy			
		Hom	MHom	Het	MHet
LINE	Gene-Dis.	89.09%	**89.39%**	89.09%	**89.42%**
	Gene-Phen.	80.12%	**80.97%**	80.13%	**80.83%**
	RNA-Dis.	75.38%	**75.71%**	49.51%	49.34%
	RNA-Gene	61.44%	**66.49%**	43.58%	**48.00%**
	RNA-Phen.	74.29%	**74.67%**	49.87%	**50.21%**
n2v	Gene-Dis.	90.43%	89.68%	90.00%	89.65%
	Gene-Phen.	82.06%	81.31%	81.76%	81.08%
	RNA-Dis.	76.28%	76.02%	50.18%	49.96%
	RNA-Gene	72.31%	**72.90%**	51.96%	**52.85%**
	RNA-Phen.	74.94%	74.73%	50.77%	50.33%
TransE	Gene-Dis.	75.51%	**80.47%**	75.53%	**80.47%**
	Gene-Phen.	64.30%	**69.84%**	64.54%	**69.84%**
	RNA-Dis.	75.78%	74.41%	84.01%	53.46%
	RNA-Gene	58.71%	**63.99%**	64.87%	**70.91%**
	RNA-Phen.	75.85%	73.14%	83.14%	55.30%

erogeneous link prediction (`Het`*, predicting both the existence and type of an edge), and their corresponding multimodal variants (*`MHom` *and* `MHet`*). DNABERT was used to embed sequences and BioBERT for descriptions. We concatenated these embeddings with those produced by LINE, node2vec (*`n2v`*), and TransE.*

Results confirm that node2vec outperforms LINE and TransE in homogeneous prediction; TransE outperforms LINE and node2vec in heterogeneous link prediction tasks when more than one edge type is present at schema-level (i.e. miRNA-gene and miRNA-disease). TransE achieves better results by disambiguating the four semantically distinct edge types between miRNAs and diseases. Finally, the multi-modal approach improves the performance in 9 out of 15 cases for the homogeneous link prediction task and in 8 cases for the heterogeneous task (highlighted in bold), with gains up to +5% in balanced accuracy observed for RNA-gene predictions in both LINE and TransE. Supplementary material reports the used parameters.

Time-stratified link prediction.

A key property introduced in RNA-KG 2.0 is the PubMed identifier of scientific articles that support the existence of relations among bio-entities (14.3*M* edges have been annotated). From these identifiers, we can extract the publication year that can be exploited for observing the content of the KG at different instants of time.

Example 9. *To show the relevance of the publication date when checking for new advancements on regulatory RNA molecules for the development of new therapies and vaccines, we extracted interactions from RNA-KG involving miRNA, gene, diseases, phenotypes, and epigenetic modifications. They are supported by 52k PubMed articles that we grouped together relying on the year of publication. Starting from the graphical distribution of articles per year reported in Fig. 10, we can observe that more than 84% of the relations were discovered after 2019, the year marking the emergence of COVID-19 and a subsequent surge in RNA-related studies.*

Time-stratified link prediction can be realized by training models with the knowledge available up to a given instant of time and then evaluate their ability to predict links discovered later. KG quality can be validated through this approach, as it is expected to support the discovery of future knowledge based on past evidence.

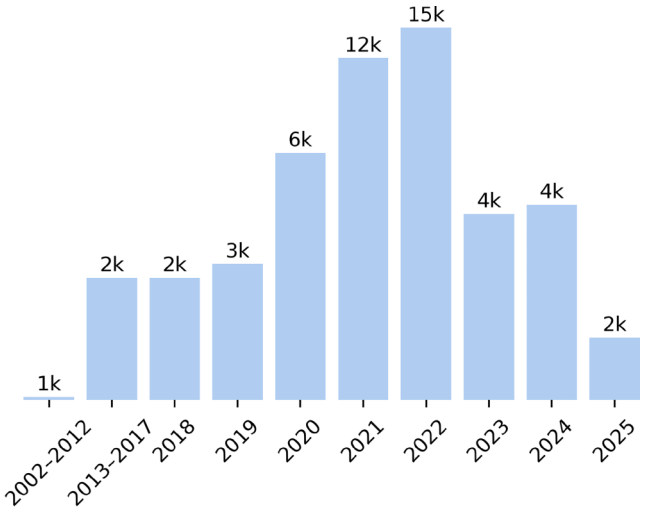


Figure 10. Distribution of articles per year in *miRNAdisease*.

Example 10. *To validate the miRNAdisease view using a time-stratified scenario, we trained a ML model on its relations supported by articles published before 2022. Interactions discovered after 2022 are used as a test set. These links represent knowledge that was not available at training time and to which we aim to assign high probability scores. The model has been conceived through node2vec embeddings coupled with a RF classifier, following our link prediction pipeline* [24].

Fig. 11 reports the distributions of probability scores for interactions involving miRNA molecules in the test set. Results show that 82% of the predicted links achieve prediction scores well above 0.5 (highlighted in red), with median scores exceeding 0.6. The boxes corresponding to years 2023 and 2024 appear wider, possibly due to a larger number of novel interactions and/or increased variability in their nature. This suggests that the KG can support meaningful predictive infer-

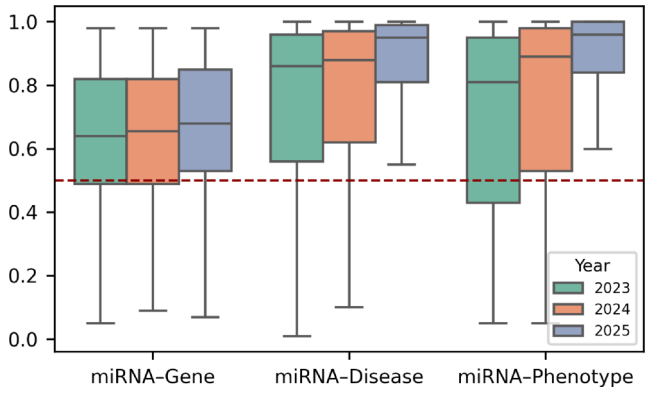


Figure 11. node2vec probabilities for relations discovered after 2022.

ence, as the model can anticipate future associations relying on prior evidence.

Hypothesis generation through RNA-KG

RNA-KG v2.0 facilitates the generation of new, testable biological hypotheses by integrating molecular interaction and sequence information, allowing researchers to identify candidate molecules, regulatory mechanisms, and disease associations for further investigation.

Example 11. *Using the API call of Example 2, we discover the existence of a siRNA sequence (https://web.mit.edu/sirna/sequences/results-1061.html) that represses* `EZR-201` *with a reported knockdown percentage of 70–80%. The presence of a siRNA targeting the transcript may contribute to hypothesis generation in two ways: (i) analyzing the siRNA sequence can guide the design or discovery of novel siRNAs or ASOs with stronger regulatory effects; (ii) using the RNAcentral 'Search' tool, this siRNA is found to share similarity with four poorly characterized human sRNAs, potentially worth further investigation. Interestingly, these sRNAs interact with several lncRNAs in RNA-KG.*

ML models trained on RNA-KG can be used to generate hypotheses by predicting novel RNA–disease associations that are not documented in RNA-KG. Articles associated with candidate RNAs can be explored through the RNAcentral 'Publications' tab to assess involvement in diseases of interest.

Example 12. *Using a node2vec+RF model trained on the miRNAdisease view, we infer that* `hsa-miR-345-5p` *is associated with ovarian cancer with a prediction score of 0.80. Although this association is not represented in RNA-KG, a researcher finds that this claim is supported by the literature using the RNAcentral 'Publications' tab with 'ovarian cancer' as filter. Experimental studies can be prioritized based on the involvement of this miRNA in ovarian cancer over other candidate miRNAs/diseases.*

Integrating RNA-KG with external -omics sources

Several types of analyses can be performed by integrating the information available in RNA-KG and other -omics data sources, leveraging the provided API. Such integration enables the combination of structural, functional, interaction-based, expression, and literature-derived facts about RNA molecules, which are often not available in a single resource.

Using the recent integration of RNA-KG relationships in the RNAcentral portal [19], a biologist interested in a specific molecule, such as the human microRNA *hsa-miR-183-5p* involved in cancer progression, can visually explore its interaction network through the RNAcentral 'Relationships' tab (highlighted in red in Fig. 12). From this view, rapid hypothesis generation can be performed without requiring API calls or programming skills.

Moreover, additional mRNA interactions from IntAct [88], conserved sequence annotations, RNA editing annotations from REDIportal, and functional GO annotations can be easily accessed and integrated with the interactions contained in RNA-KG. Cross-referencing these annotations compensates for information intentionally omitted from RNA-KG to reduce the graph size (e.g. A-to-I editing events would dominate the graph structure if represented as relationships), allowing their retrieval 'on-the-fly' only when actually needed.

Furthermore, the RNAcentral 'Search' tool can be used to retrieve candidate ASO, siRNA, and miRNA sequences, expanding sets with conserved homologs or natural counterparts similar to synthetic inputs. By cross-referencing these sequences with RNA-KG relationships, researchers can assess potential participation in regulatory networks, interactions with disease targets, or shared functional annotations, helping prioritize candidates for experimental validation or therapeutic design.

Finally, additional RNAcentral functionalities (e.g. the new 'Gene' tab, expression profiles across tissues from Expression Atlas [27] accessible within the 'Expression' tab, LitSumm [89], R2DT [90]) and complementary prediction tools (e.g. DIANA-microT) can be exploited to further enrich the characterization of RNA molecules, providing insights into family relationships, structural and expression features, and predicted binding sites and binding types (e.g. 6mer, 9mer). These data can also be incorporated into multimodal ML models to complement interaction- and property-based evidence, similarly to what has been done in: (*i*) MMnc [91], which proposes a deep-learning framework that classifies non-coding RNAs into functional groups by leveraging sequences, secondary structures, and expression profiles; (*ii*) MolFM (Multimodal Molecular Foundation Model [92]), which integrates molecular structure, biomedical literature, and KG data through a cross-modal attention mechanism for performing property prediction, drug–target interaction prediction, and molecule–text retrieval; and (*iii*) M2REMAP [93], which embeds chemical structures, patient records, and drug–disease relationships to prioritize molecule–disease associations.

Example 13. *Suppose a biologist wishes to extract a subnetwork of lncRNA–miRNA interactions associated with cancer-related processes and identify a molecular relationship between the lncRNA* `TSIX` *and the miRNA* `hsa-miR-155-5p`*. This interaction, imported from LncBase [94] and characterized by a DIANA-microT score of 0.82, indicates a high-confidence for playing a role in post-transcriptional regulation relevant to oncogenic mechanisms, and thus warrants further experimental validation. To enrich this relationship with contextual biological evidence, complementary data can be retrieved from external sources. For example, RNAcentral provides conserved sequence motifs for both molecules and a textual summary for* `TSIX`*, while REDIportal characterizes RNA editing events occurring on the sequence of* `TSIX` *and that may affect the binding affinity between them.*

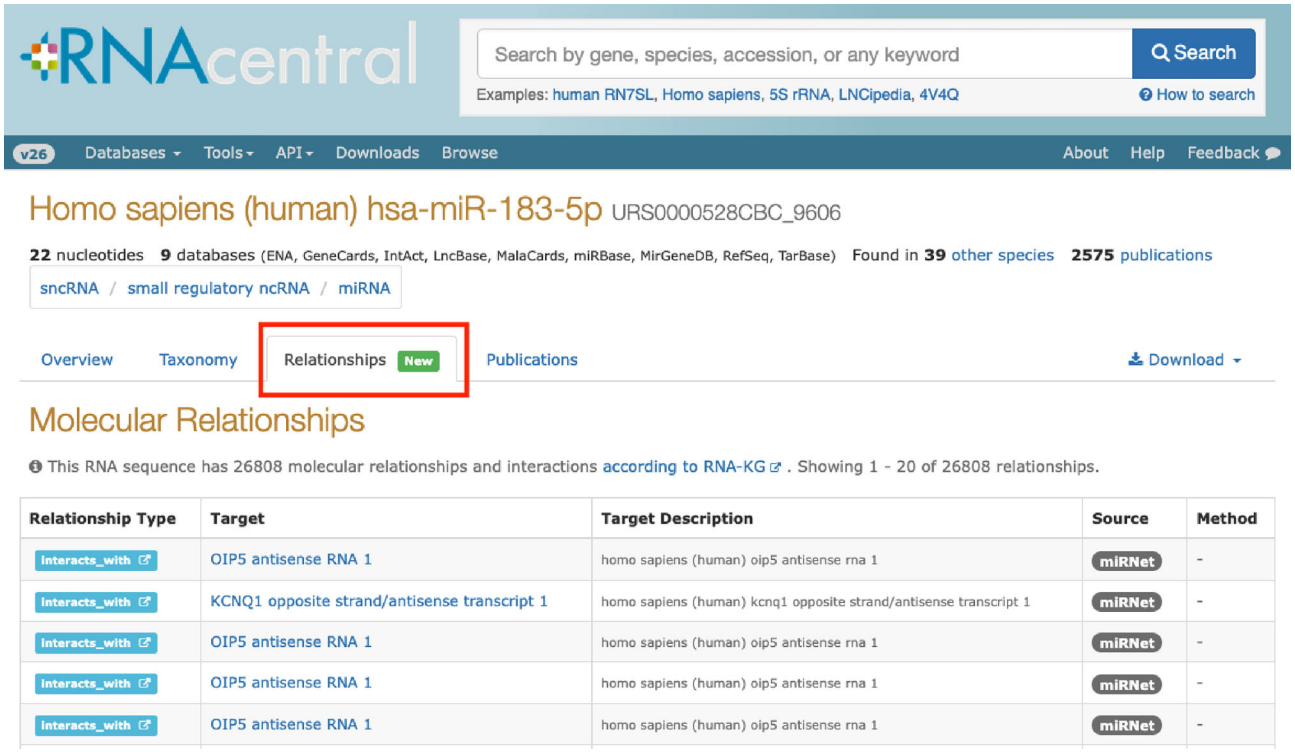


Figure 12. RNAcentral 'Relationships' tab for *hsa-miR-183-5p*.

Fig. 13 shows the pipeline for integrating this complementary evidence within the subgraph extracted from RNA-KG. Motifs and editing events are modeled as nodes connected to `TSIX` *and* `hsa-miR-155-5p` *through the relationship* characteristic of part of *(annotated with the nucleotide interval where the motif occurs e.g. '1–23') and* involved in *(annotated with the positions of editing events e.g. '1828', '1840'). Moreover, introducing a property to the* `TSIX` *node with the text* ' `TSIX` is a non-coding RNA that plays a crucial role in the regulation of X chromosome inactivation, particularly by repressing the expression of its antisense transcript XIST. It is involved in a complex regulatory network where its expression can be influenced by factors such as REX1 and OCT4.', *reinforces the hypothesis that* `TSIX` *participates in complex post-transcriptional networks and interact with specific miRNAs.*

The integrated subgraph can be generated 'on-the-fly' and leveraged to accelerate hypothesis generation by assessing whether interactions are consistent with known regulatory mechanisms such as the sponge-like function of lncRNAs in miRNA-mediated regulation (ceRNA). This supports the prioritization of the most plausible candidates for experimental validation. Finally, the enriched subgraph can serve as input for downstream ML models that encode topological, sequence-, text-, and event-based features, injecting further context into RNA–RNA interaction prediction and improving explainability of inferred relationships.

Discussion

In this paper, we presented the new characteristics of the enhanced version of RNA-KG. It now contains properties associated with bio-entities and their interactions that can be exploited for various kinds of analysis and research activities. We described various use cases that leverage the new information contained in this version of the system. The KG is publicly accessible, can be easily queried, and is integrated in the RNAcentral portal.

We are currently working on enhancing RNA-KG in different directions. While the current version focuses on human molecules, entities and relationships from other species should be included, along with additional information extracted from external sources. This expansion will increase the RNA-KG coverage and enable comparative and cross-species analyses.

The use of automatic techniques to incorporate additional facts into RNA-KG may propagate errors already present in the original sources. In future work, we plan to adopt anomaly detection techniques [95] for KGs to prevent the introduction of potential false-positive relationships. Moreover, RNA-KG is static while biological networks are context-dependent and relationships among molecules often hold only within specific subsystems (e.g. n-ary interactions such as ceRNA mechanisms) or temporal windows. Incorporating temporal dimensions and contextual constraints would allow modeling dynamic changes in RNA interactions. To further enhance relationship representation, a hypergraph model should be used.

The integration of external -omics or predictive resources may require additional normalization and mapping efforts. Although many lookup tables are provided to support this process, they are static and must be regularly updated to reflect evolving identifiers and naming conventions. The development of an RNA ontology could mitigate this limitation by providing an up-to-date mapping framework.

The presence of an ontology is of paramount importance for the description of the characteristics of coding and non-coding RNA molecules. However, the current tentative proposals do not take the heterogeneity of identification schemes across RNA sources into account, do not provide mapping

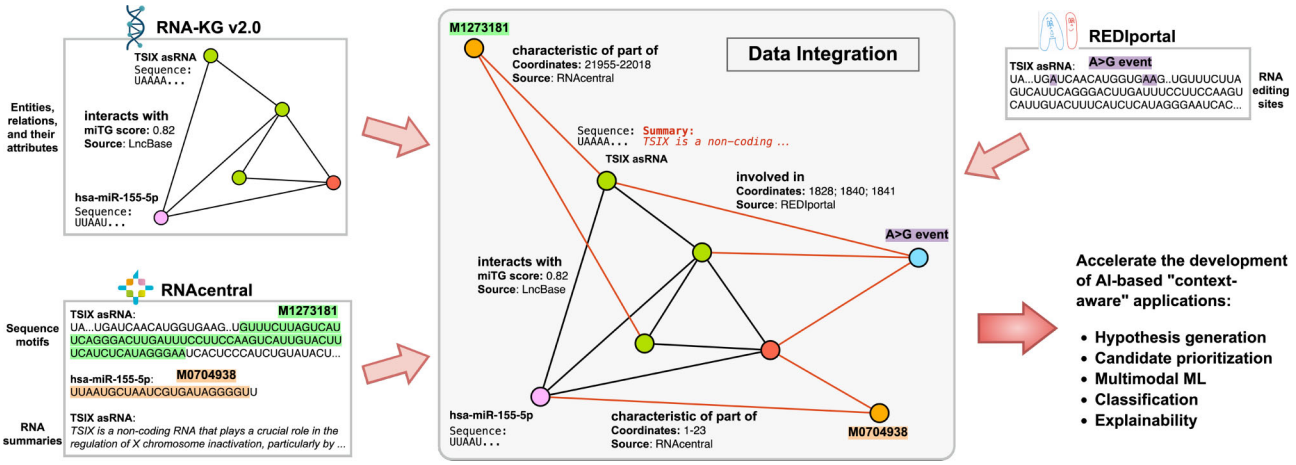


Figure 13. Example of data integration pipeline involving relationships from RNA-KG v2.0.

rules between identification systems, and do not include properties that characterize RNA molecules (e.g. sequences, structures, and descriptions). Moreover, they fail to capture recent advancements in RNA research. The Non-Coding RNA Ontology (NCRO [96]) was published over eight years ago and focuses on the classification of non-coding RNA molecules with an emphasis on representation of the over 2*k* human stem-loop miRNAs. However, cross-references are not up-to-date and relevant properties (like the sequence) are not provided. NCRO also lacks representation of other ncRNA molecules (e.g. lncRNA, circRNA) now recognized as pivotal in therapeutic and diagnostic contexts. The RNA Ontology (RNAO [97]) was published more than fifteen years ago and focuses on 3D RNA structures, including classes about biochemical and molecular types but does not include RNA sequences, limiting its utility in sequence-specific applications.

For these reasons, we would like to develop a new ontology that leverages the knowledge contained in RNA-KG v2.0 and complements existing RNA sequence repositories such as RNAcentral and Ensembl by providing an ontological representation of RNA transcripts. The new 'Gene' feature recently introduced in RNAcentral already moves in this direction, as it clusters similar transcripts into gene-level entries with their own identifier. Such structured information could serve as a foundation for building the new RNA ontology and better annotate many experiments about RNA interactions that operate at the gene level rather than the transcript level.

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Author contributions: E.C. and M.M. conceived the experiment(s), E.C. conducted the experiment(s), E.C. and P.P. developed the RNA-KG web portal, E.C. and M.M. analysed the results. E.C., P.P., and M.M. wrote and reviewed the manuscript.

Supplementary data

Supplementary data is available at NAR Genomics & Bioinformatics online.

Conflict of interest

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Data availability

The new release of RNA-KG is stored as a property graph database. A web portal is available at https://RNA-KG.biodata.di.unimi.it and provides several facilities for working with RNA-KG v2.0, and a public Neo4j query endpoint has been realized and made available at https://neo4j.biodata.di.unimi.it with username and password *rnakgv20*. Data and code for reproducing experiments are available in Zenodo and GitHub at the links: https://doi.org/10.5281/zenodo.10078876; https://github.com/AnacletoLAB/RNA-KG. The GitHub repository provides Python scripts and Jupyter notebooks (stored in the `notebooks` directory) to process from scratch the 91 linked open data repositories and ontologies that compose the current release of RNA-KG. These scripts can be applied to process similar sources, and the GitHub repository can be used to suggest new data providers to our team via the `Issues` tab. Code is available for each of the following KG construction steps: (*i*) Download, clean, and merge the 11 ontologies describing RNA-KG v2.0 schema (ontologies are processed according to PheKnowLator's implementation of OWL-NETS [12, 66]); (*ii*) define and generate lookup tables for normalizing entities according to standard identi-

fiers; (*iii*) process entities, relationships, and their properties from the 80 linked open data repositories. Finally, a bash command is provided to import the generated files (*nodes.csv* and *edges.csv*) into a Neo4j server storage.

The Zenodo repository provides: (*i*) descriptive statistics about RNA-KG v2.0, (*ii*) a copy of the intermediate *nodes.csv* and *edges.csv* files to reduce the need for re-running the entire pipeline, (*iii*) a Neo4j dump of the full database, and (*iv*) a *resources.zip* archive containing additional intermediate files relevant for constructing the KG. The latter may be particularly useful for reusing pre-processed lookup tables (e.g. ones involving RNAcentral or OBO ontologies identifiers) and building other biomedical KGs from scratch.

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